Advancements in Carbon Dioxide Utilisation: Pathways to Sustainable Chemicals, Fuels, and Materials for a Net-Zero Future

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ABSTRACT

The increasing concentration of atmospheric carbon dioxide (CO_2) has intensified global efforts to develop Carbon Capture, Storage and Utilisation (CCSU) technologies. This review examines key CO_2 utilisation routes, including chemical transformation, mineralisation and biological transformation. Major CO_2 products such as methanol, urea, formic acid, dimethyl carbonate, and polycarbonates, are discussed with respect to reaction mechanisms, catalysts, industrial processes, and applications. Emerging uses in pharmaceutical and biotechnological industries, where CO_2 is used as a green reagent and solvent, are also discussed. Recent trends indicate emerging interest in industry and the adoption of technology, but issues such as high energy demand, catalyst constraints, and economic limitations remain a point of concern. To improve efficiency and scalability, there is a need to integrate renewable energy sources with sophisticated catalytic systems. Overall, CO_2 utilisation presents a promising pathway toward a circular carbon economy and supports global efforts to achieve net-zero emissions and increase possibilities of continued use of fossil fuels such as coal, petroleum and natural gas. This paper highlights recent advances (2024–2026) and identifies key challenges

for large-scale implementation.

KEYWORDS: CO_2 utilisation; CCSU; carbon capture; green chemistry; sustainable fuels

1. Introduction

Carbon dioxide (CO_2), long considered a major greenhouse gas contributing to climate change, is increasingly being recognized as an important carbon feedstock in the chemical, energy, and materials industries (Aresta et al., 2014; Centi & Perathoner, 2009; Sharma, 2021; Kim et al., 2022). Emissions of CO_2 worldwide exceed 35 billion tonnes per year (IEA, 2024), most of which are from power generation, steel, cement, chemicals, and transport sectors (IEA, 2024). As long as fossil fuels remain the main source of energy, there is a dire need to develop policies to reduce CO_2 levels whilst simultaneously expanding industrial production (Olah et al., 2011; Aresta et al., 2014; Sharma,2023). Carbon Capture, Storage and Utilisation (CCSU) methods offer a two-fold advantage by reducing emissions and enabling the continued use of fossil fuels by converting CO_2 into value-added products. This review repositions CO_2 from an environmental liability to a strategic carbon resource (Keith et al., 2018). The high thermodynamic stability of CO_2 ($\Delta G°f = -394.4$ kJ/mol) makes its activation kinetically challenging (Lackner, 2003; Sharma, 2010), yet catalysis, electrochemistry, and biotechnology techniques help overcome these barriers. One example is chemical changes, such as dry reforming of methane and CO_2 to produce syngas (H_2 + CO), which can form hydrocarbons via Fischer-Tropsch synthesis (Pakhare & Spivey, 2014). Bioconversion uses microorganisms such as Moorella thermoacetica to convert organic acids, which can then be mineralized into construction materials using fly ash or steel slag in a circular economy (Bagotia et al., 2018). Several techniques for the chemical and

bioconversion of CO_2 have been suggested recently along with the directions for future research work for developing circular economy using CO_2 (Sharma, 2021; Sharma ,2023). The current trends (2024-2026) indicate the pace as the currently existing global market of CO_2 utilisation is USD 4.57 billion in 2024 and will become USD 11.15 billion in 2032 at a CAGR of 13.6% (IEA, 2024), whereas the larger markets of utilizing carbon capture undergo expansions of USD 3.5 billion in 2026 and USD 51.6 billion in 2050 (IPCC,2023), Some of the driving aspects are growing global climate policies, where more than 474 CCSU projects have been announced, aiming to reach the eight hundred and fifteen million tonnes per annum by 2030 (Global CCS Institute, 2024), and investment such as the United States Inflation Reduction Act allocating one point seven billion to capture demonstrations. The Icelandic Mammoth DAC i.e., Direct Air Capture (launched in 2026, capturing 36,000 tonnes/year) and the DAC unit at Aramco (12 tonnes/year in 2025) are among the first steps. This need is further supported by the fact that 53,970 publications on CCSU in 2024 are 11.4% higher than the previous year (Global CCS Institute, 2024). This review compiles findings from chemical pathways, industrial case studies, and new trends to highlight the role of CO_2 in advancing the low-carbon economy. Since there are extensive reserves of coal, a cheaper source of energy available in the world, therefore, there is a grave need for exploring economical possibilities of their continued use for power generation. That would become possible only if the CCSU technologies are economically established.

2. Material and Methods

This study is a review-based analysis of carbon dioxide (CO_2) utilisation technologies. Relevant literature was collected from peer-reviewed journals, books, and industry reports focusing on chemical, biological, and industrial applications of

CO_2. The sources were chosen for relevance, credibility, and recency, and developments that occurred between 2010 and 2026 were given special attention. The review synthesises information on reaction pathways, catalysts, industrial processes, and emerging applications to provide a comprehensive understanding of CO_2 utilisation.

3. Results and Discussion

(1) Objective and Scope

This review aims to analyse key pathways for CO_2 utilisation, including hydrogenation, carboxylation, and reforming processes. It assesses the industrial processes, catalysts, and reaction mechanisms of transforming CO_2 to value-added products. The research also contrasts the leading CO_2-based chemicals, including methanol, urea, formic acid, dimethyl carbonate, metal carbonates and polycarbonates, in terms of their sustainability and industrial applicability. Recent developments and future trends are also encompassed in the scope, as well as emerging applications in pharmaceuticals, biotechnology and construction materials.

(2) CO_2 as a Resource

Various industries emit large amounts of CO_2, and it is inexpensive and non-toxic, making it an appealing chemical feedstock. Nonetheless, it needs to be activated by catalysts, heat, or renewable electricity to achieve thermodynamic stability (Alper & Orhan, 2017). There could also be a need for the costly hydrogen under pressure to hydrogenate CO_2 uneconomically. Efforts are required to avoid using hydrogen to the extent possible though it is not possible always and thus this may be done to the extent possible economically.

CO_2 utilisation contributes to sustainability by reducing greenhouse gas emissions, replacing fossil-derived chemicals with CO_2-based alternatives, and enabling circular carbon economy models such as CO_2-based gasification processes for syngas production. It helps in enabling circular carbon models, e.g., CO_2 and coal gasification (Dhawan et al., Pakhare & Spivey, 2014; Sharma and Giri, 2016; Dhawan et al., 2020), to produce syngas (Centi & Perathoner, 2009).

According to historical reviews, CO_2 was considered an inert gas (Peters et al., 2011; Quadrelli et al., 2011); however, it is now widely used in urea production (on an industrial scale) and in biofuels with potential. New trends are moving more rapidly: the EU launched an Innovation Fund worth 1.5 billion to support CCSU projects, and China, in its Five-Year Plan, is interested in CCSU, increasing DAC and utilisation (Global CCS Institute, 2024). Indian Government has committed Rs. 20,000 crores towards efforts on CCSU for the 2026 -2027 budget (https://www.downtoearth.org.in/climate-change/indias-rs-20000-crore-CCSU-push-what-the-union-budgets-carbon-capture-bet-really-means, accessed on April 14, 2026).

According to the Global Carbon Budget, global CO_2 emissions were projected to reach 41.6 billion tonnes by 2024, underscoring the need for CCU (IEA,2024) and these have understandably increased further in 2026.

Carbon dioxide is thermodynamically stable due to its linear molecular structure and strong C=O double bonds (~750 kJ/mol bond energy), with a standard Gibbs free energy of formation of $\Delta Gf^{\circ}(CO_2) = -394.4$ kJ/mol. This high stability implies a large activation energy barrier, kinetic inertness under ambient

conditions, and the need for external energy input. The primary energy penalty sources include hydrogen production (especially if not green), high pressure (20–100 bar), elevated temperatures (200–600°C), and catalyst regeneration.

(3) CO_2 Utilisation Pathways

CO_2 utilisation can be broadly classified into chemical conversion, mineralisation, and biological transformation; shown in Figure 1.

Figure 1. Classification of CO_2 utilisation pathways.

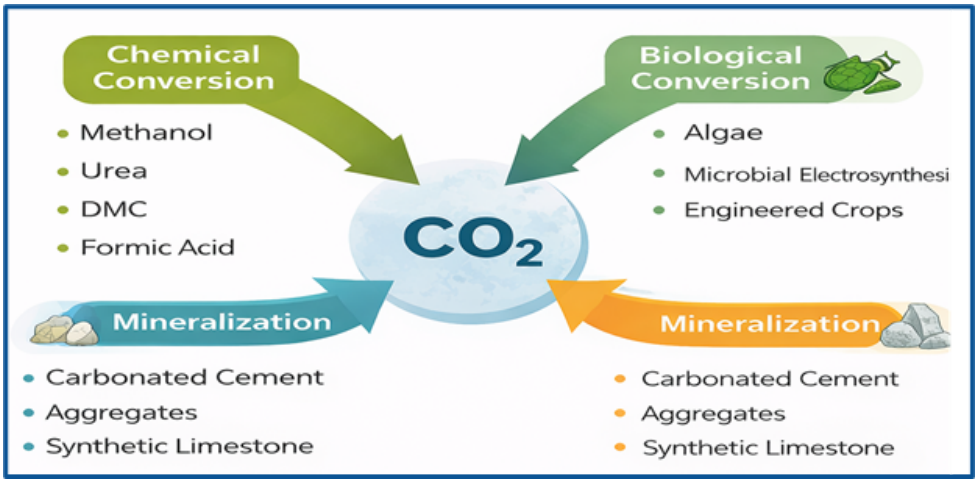


Image description: Diagram illustrating chemical, biological, and mineralisation pathways of CO_2 leading to fuels, chemicals, and materials.

Chemical conversions may take the form of hydrogenation, such as $CO_2 + H_2$ producing methanol (Pontzen et al., 2011), carboxylation, cycloaddition, or electrochemical reduction (Jhong et al., 2013; Nitopi et al., 2019; Hori, 2008). Mineralisation involves reactions of CO_2 with metal oxides or industrial waste materials to form stable carbonates (Bagotia et al., 2018, Dhawan et al., 2019). Mineralisation provides permanent carbon storage, but its economic viability depends on material availability and logistics. Lime, coal fly ash, steel slag or silicate ores are readily available. Besides lime conversion to $CaCO_3$, other

conversions may require severe conditions for conversion and there further research work is warranted.

Biological transformation involves using microbial and algal systems to convert CO_2 into biofuels, organic compounds, and bioplastics (Global CCS Institute, 2024). Gautam et al (2015) have reviewed the research work on utilization of CO_2 for the growth of algae. The photosynthetic efficiency of algal systems is higher in comparison to that of land based plants. However, there is a need to economise the algal biorefineries (Tyagi et al., 2014; Gautam et al., 2015; Manchanda et al., 2021). Though Rubisco is a lethargic enzyme still photosyntheses offers a potential natural CCSU process and there is a possibility of utilizing lignocellulosic as well as algal biorefineries for the production of biofuels and utilizing CO2. Jahid et al. (2018) have reported the production of bioethanol using fruit wastes through chemical as well as biochemical hydrolysis processes. Potential technologies for the production of chemicals, fuels and value-added chemicals from biomass wastes has been reviewed by Sharma (2015).

Table 1. Comparative techno-economic analysis of CO_2 utilisation pathways.

Pathway	TRL	Energy Input	Carbon Permanence	Scalability	Main Limitation
Hydrogenation (Methanol)	8-9	High (H2 dependent)	Medium	Industrial	H2 cost
Electrochemical	5-7	Electricity based	Low-medium	Pilot	Electrode Stability
Mineralization	7-8	Moderate	Very high	Scalable	Material Logistics

Bioconversion	4-6	Low medium	Low	Experimental	Slow rates

4. Chemical Conversion of CO_2

The activation of stable CO_2 is carried out through several chemical means. Green hydrogenation using H_2 occurs at 200–300°C and 20–100 bar using Cu/ZnO catalysts (Goeppert et al., 2014; Bowker et al., 2019; Sabatier, 1912, Kim et al., 2022). Another method is formic acid reduction by renewable electricity using Sn electrodes, while sunlight-based photocatalysis often utilizes TiO_2-based materials (Whipple & Kenis, 2010). In the field of polymers, activation involves carboxylation/cycloaddition (Kätelhön et al., 2019), and metal-organic frameworks (MOFs) are employed to enhance selectivity in the cycloaddition reaction of epoxides (Elhenawy et al., 2020).

Table 2. Comparison of Chemical Conversion Methods (Updated with 2024-2026 Data)

Method	Energy Source	Products	Challenges	Recent Advances
Hydrogenation	Heat/ H2	Methanol, Formic acid	H2 cost	KS-21 solvent; 88% CO_2 conversion
Electrochemical	Electricity	Formic acid, CO_2, Ethanol	Electrode Durability	Oxylus direct methanol: >80% selectivity
Dry Reforming	Heat	Syngas	Catalyst Deactivation	Ni-Fe alloys: BRM with 72.5% CO_2 conversion

Dry reforming of methane using CO_2 ($CH4 + CO_2 \rightarrow 2CO + 2H_2$) involves Ni-based catalysts, and Fe promoters minimize the coke formation (Pakhare & Spivey, 2014; Manabayeva et al. ,2023). Several bimetallic and trimetallic catalysts along with Ni are being utilized. There is a formation of carbon nanomaterials which may be recovered to activate the catalysts used.Tri-reforming stabilizes the process by adding H_2O and O_2 (Swirk et al., 2017). Recent innovations include solvents such as KS-21, which has been used for large-scale capture, combined MOF-sorbents with improved efficiency, and electrochemical CO_2-to-ethanol conversion.

Methane bi-reforming (BRM) has shown promise, with conversion rates exceeding 80% using CaO catalysts. Research work on bireforming of methane has been reviewed by Chan et al. (2025).

5. Industrially Relevant CO_2-Derived Chemicals

(1) Methanol

Reaction: $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$ ($\Delta H = -49$ kJ/mol).

Catalysts: $Cu/ZnO/Al_2O_3$ (standard industrial process).

Procedure: feed (CO_2) (flue gas/DAC) and green H_2 (generated in the electrolysis); reaction conditions : 200–300°C, 20-100 bar; pathway; $CO_2 \rightarrow$ formate $\rightarrow$ formaldehyde $\rightarrow$ methoxy $\rightarrow$ methanol; distillation; recycle unreacted gases. (Goeppert et al., 2014; Behrens et al., 2012, Kim et al., 2022).

It is used as a fuel additive and as a precursor to formaldehyde, acetic acid, and olefins (in the MTO process). Ethanol has potential as a fuel derived from CO_2, and pilots such as Oxylus' electrochemical conversion are scheduled for 2026.

Figure 2 shows the process of conversion of CO_2 to methanol in industries.

Figure 2. Industrial CO_2-to-methanol production process.



Image description: Process diagram showing CO_2 and hydrogen input into a reactor for methanol production under controlled temperature and pressure.

Although methanol synthesis from CO_2 is industrially established, its sustainability depends heavily on the availability of green hydrogen, which remains a key limitation.

(2) Urea

Reaction: $2NH_3 + CO_2 \rightarrow NH_4NH_2COO \rightarrow CO(NH_2)_2 + H_2O$

Process: $CO_2 + NH_3$ gives ammonium carbamate (140-175 bar); perform dehydration and recycle NH_3.

Its uses are fertilizer, plastics, and resin. Urea production is one of the largest industrial applications of CO_2 utilisation globally.

(3) Formic Acid

Routes: (1) $CO_2 + H_2 \rightarrow HCOOH$;

(2) Electrochemical: $CO_2 + 2e^- + 2H^+ \rightarrow HCOOH$.

Catalysts include: Homogeneous (Ru, Ir complexes) and Heterogeneous (Sn, Bi, In, Pb electrodes). Uses: Textile/leather treatment, H2 storage. The

electrochemical paths are best used in distributed production (De Luna et al., 2019), More than 75% yield with DAC reported by Bajracharya et al., (2017). Electrochemical CO_2 reduction offers high selectivity and flexibility; however, challenges related to electrode stability and energy efficiency limit its large-scale application.

(4) Dimethyl Carbonate (DMC)

Reaction: $CO_2 + 2CH_3OH \rightarrow (CH_3O)_2CO + H_2O$

Challenges include thermodynamic constraints, the need for water removal, and the requirement for strong catalysts.

It has uses as a green solvent, a fuel additive, and a phosgene replacement.

(5) Polyols & Cyclic Carbonates

Reactions: Epoxide + CO_2 → cyclic carbonate; copolymerization to polycarbonate polyols (Ye et al., 2019). The catalysts are: Zn, Co metal complexes; ionic liquids; organocatalysts.

Uses include polyurethanes, battery electrolytes, and engineering plastics. 40-50 wt% CO_2 locked (Artz et al., 2018); Graphene composite of CO_2 (Bagotia et al., 2018, Artz et al., 2018).

6. Graphene and Carbon Nanomaterials

Carbon dioxide can also be utilized for the production of advanced carbon nanomaterials, such as, graphene and carbon nanotubes, which have significant industrial applications in space crafts, transport vehicles, sports, defence, electronic etc. (Bagotia et al., 2018a).

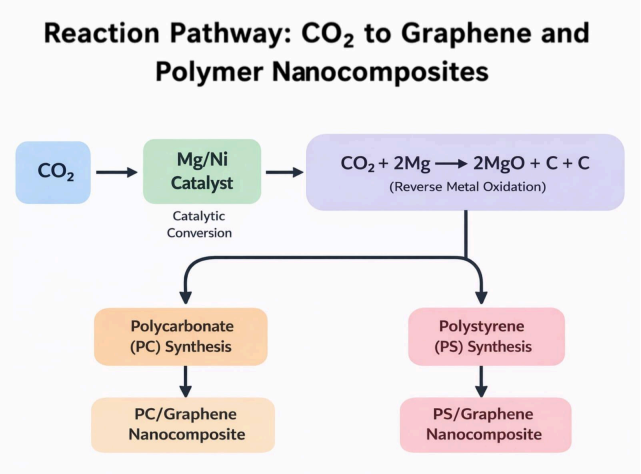
In fact , solid CO2 can be utilized by using burning Mg ribbon to produce graphene. Bagotia et al. (2018b) have used Ni and Zn as promoters in this reaction to produce

graphene having better characteristics and in higher yields. Graphene can be converted to carbon nanotubes (Bagotia et al., 2018b). Recent studies highlight the potential of CO_2 as a carbon source for synthesizing high-value nanostructured materials. CO_2 can also be utilized for producing polycarbonates (Bagotia et al., 2018a). In fact, Bagotia et al. (2019) have utilized graphene and multiwalled carbon nanotubes for producing polycarbonate/ ethylene methyl acrylate nanocomposites having enhanced mechanical, electrical and EMI shielding properties. Graphene and carbon nanotubes can be produced from CO_2 using Mg reduction, and Bagotia et al. (2018a) have reported the enhancement in the production of graphene using Ni and Zn as promoters. Here Ni was found to be a better promoter than Zn.

In fact, the use of graphene-based nanocomposites i.e., polycarbonate/ ethylene methyl acrylate nanocomposites showed better electrical conductivity, thermal conductivity, and EMI shielding properties.

Sharma (2023) has reported the production potential of carbon nanomaterials using cheaper feedstocks, such as, coal, CO_2, waste plastics, non-edible vegetable oils etc. Kim et al. (2022) and Yousuf et al. (2026) have reviewed the research work on the industrial production of chemicals from the CO_2 and the recent advances in the CCSU. The utilization of CO2 for biofuels and petrochemical production is widely known now. However, present authors have focused on the utilization of CO_2 by the pharmaceutical industries. Figure 3 shows the process of production of graphene from CO2 and utilization of graphene and carbon nanotubes for the production of polycarbonate and polystyrene.

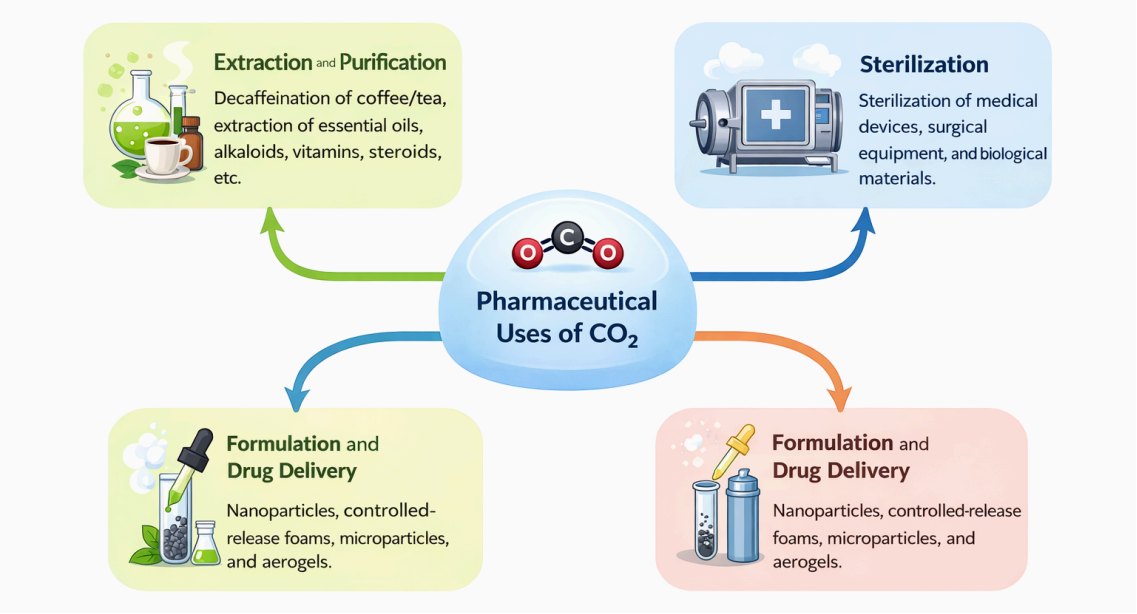
Figure 3. Schematic representation of CO_2 conversion into graphene using Mg/Ni catalysts and its subsequent utilization for the synthesis of polycarbonate and polystyrene-based nanocomposites.



7. Uses of CO_2 in the Pharmaceutical Industry

In contemporary pharmaceutical processes, CO_2 is important not only as a chemical reagent but also as a green solvent, extraction medium, sterilizing agent, and formulation component (Elhenawy et al., 2020). Its non-flammable, non-toxic, cost-effective, and non-polluting characteristics have made it a choice alternative to harmful organic solvents. Figure 4 shows the different Pharmaceutical uses of CO_2.

Figure 4: Pharmaceutical uses of CO_2



Important CO_2-Based Reactions in Drug Synthesis :

(1) *Carboxylation Reactions*

One of the most basic reactions of CO_2 in the synthesis of organic compounds is carboxylation.

General Reaction: R-MgX + CO_2 - R-COOH.

In this reaction, organometallic reagents such as Grignard reagents (R-MgX) or organolithium compounds react with CO_2. The resulting magnesium carboxylate is acidified to get carboxylic acids.

Medical Significance: These carboxylic acids are key intermediates in the synthesis of NSAIDs (e.g., ibuprofen-type molecules), antibiotics, antifungals, and β-lactam drugs. This reaction enables the efficient synthesis of CO_2, which is a cost-effective source of carbon.

(2) *Carbamate and Urea Formation*

CO_2 readily reacts with amines to form carbamates and urea.

General Reaction: 2 RNH_2 + CO_2 - RNH-CO-NHR + H_2O.

Pharmaceutical Significance: Antimalarial drugs, antibiotics, antiretroviral drugs, antidiabetic drugs, and enzymes are produced. Examples include prodrug formation (carbamates shield amino groups) and drug stability (in fact, the urea derivatives enhance bioavailability).

(3) *Reactions in Supercritical CO_2 ($scCO_2$)*

Supercritical CO_2 ($scCO_2$) above 31.1°C and 73.8 bar is a dual solvent, both liquid and gas, and is considered a green solvent.

Some of the applications in pharmaceutical synthesis include: $scCO_2$ -assisted organic reactions, such as hydrogenation, oxidation, polymerization; particle formation, crystallization, or polymorph control to achieve solubility/stability; and many others.

Its Advantages include: Removing harmful organic solvents, processing at low temperatures to maintain heat-sensitive molecules, and high diffusivity, which permits faster, cleaner reactions.

8.Pharmaceutical Uses of CO_2

(1) *Extraction and Purification*

CO_2 (at supercritical conditions) has numerous applications in the extraction of caffeine (decaffeination of coffee/tea) and essential oils, and is also used for the extraction of alkaloids, vitamins, and steroids.

$scCO_2$ is a better choice because it produces no solvent residues (enhancing drug purity), is non-toxic and GRAS, and can be used to selectively extract compounds by adjusting pressure and temperature.

(2) *Sterilization*

Supercritical CO_2 is an efficient sterilization medium for implantable medical devices, surgical equipment, biological materials, and temperature-sensitive drug formulations. Advantages over Heat or Radiation: Non-destructive sterilization; does not degrade polymeric materials or biomolecules; leaves residues of microbes or biomolecules.

(3) *Formulation and Drug Delivery*

CO_2 is used to make porous structures, microparticles, foams for controlled drug release, nano- and microencapsulation, tissue engineering materials, and aerogels.

Mechanisms: The fast growth of the supercritical solutions; foaming under the influence of CO_2; displacement of the solutions.

Its Applications include the production of micronized drug powders with improved dissolution rates and CO_2-expanded solutions for injectable or transdermal formulations.

(4) pH Control in Fermentation Processes

In the production of biopharmaceuticals, CO_2 helps maintain optimal pH in fermenters, which is then bubbled into microbial or cell culture systems. The control of pH is essential for the production of antibiotics, enzymes, monoclonal antibodies, and therapeutic proteins.

(5) CO_2 as an Environmentally Friendly Solvent

CO_2 is preferred as it is Non-flammable, chemically inert, and recyclable. Applications: Chromatography; particle engineering; drying processes; and pharmaceutical green chemistry. CO_2 enables safer, more sustainable production processes.

9. Application of CO_2 in Biotechnology

In Microbial electrosynthesis, CO_2 is reduced to acetate/ethanol with the assistance of *Moorella thermoacetica* (Rabaey & Rozendal, 2010). Algal systems: Biodiesel, proteins. Agricultural: Modified rice/Rubisco engineered to produce more yields/capture CO_2 .

Fixation by chemoautotrophic bacteria

In fact, the chemoautotrophic bacteria such as Acidithiobacillus ferrooxidans, Leptospirrulum ferrooxidans, Acidithiocillus thiooxidans, Sulfolobus acidocaldarius etc. use CO2 as the carbon source instead of carbohydrates. These microbes have been exploited in the metallurgical operations of Cu, A, U and pyritic ores. The use of these microbes in the biodesulfurization of coal has also been made (Sharma, 2021, Sharma,2023). Sharma (2021) has suggested the use of these bacteria for the

CCSU also (Sharma and Sharma, 2010). In fact, Sharma (2023) has suggested the utilization of CO2 for reaction with the side products produced during the boio desulfurization of petroleum.Sharma (2021) has also recommended several biotechnological approaches for the utilization of CO_2, one of these is to improve the efficiency of carbon dioxide fixation by all the six differeny cycles. The efficiency of lethargic rubisco enzyme should also also be improved.

10. Mineralization and Construction Materials

CO_2 + waste (fly ash, slag) yields green concrete/aggregates carbonates (Dhawan et al., 2020) which provides long-term carbon storage. Recent developments: WACKER's recovery in silicon production (IEA,2024); Mammoth DAC integration with mineralization (IPCC,2023).

- Atom Economy: Note that the reaction R-MgX + CO_2 - R-COOH has high atom economy because nearly every atom of the CO_2 molecule is incorporated into the final drug intermediate.
- E-Factor (Environmental Factor): Using $scCO_2$ as a solvent in extraction (e.g., decaffeination) drastically reduces the E-factor by eliminating hazardous organic waste like dichloromethane.

11. Sustainability, Challenges, & Future Prospects

Although the use of CO_2 has enormous potential to transform the concept of a circular carbon economy, there is significant complexity surrounding its adoption across the technical, economic, regulatory, and environmental fronts. At the same time, new vistas arise from technological advancements, policy innovations, and market dynamics, offering a way out of these hurdles. It is a section that summarizes the critical issues and opportunities based on recent analyses (2024-2026) and focuses on sustainable scaling approaches.

(1) ***Technical Challenges***

The thermodynamic stability of CO_2 requires significant energy to be activated, leading to high operating costs and energy penalties in processes such as hydrogenation and electrochemical reduction. The capture rate of existing technologies typically reaches 90 %, so to reach 98 % or more, which would qualify as net-zero, larger equipment would be used, new steps in the process would be needed, and the energy consumption per tonne of CO_2 captured would increase, which in turn becomes a factor in inflating the costs of the units. Poisoning and catalyst degradation, especially in dry reforming, are ongoing problems, and coke formation reduces efficiency (Manabayeva et al. ,2023). In the case of direct air capture (DAC) and utilisation, the technology is not yet sufficiently advanced, and it is difficult to scale low-concentration CO_2 extraction and combine it with renewable energy sources. Recent research emphasizes that the innovations in materials, such as advanced membranes and sorbents (Global CCS Institute, 2024), will be required to reduce energy requirements by 20-30% (D'Alessandro et al., 2010).

(2) ***Economic and Market Problems.***

The substantial capital and operating costs are very restrictive, and the cost of capturing CO_2 ranges from $40 to $ 120 per tonne, which is above current carbon prices and, in an unsubsidized business scenario, makes it difficult to cover the expense (IPCC ,2023). Next-wave technologies pass through "valleys of death," a lengthy development cycle, high capital expenditure, and untested business models that would discourage more traditional venture capital (World Economic Forum, 2025). The issue of fragmented markets and a lack of regular demand for CO_2-driven products also does not help the process of

commercialization, as in the chemicals industry, where the utilisation route, such as CO_2-to-methanol, would require a 14% cost reduction to compete by 2030 (Wood Mackenzie, 2026). There are also bottlenecks in monitoring, reporting, and verification (MRV) of utilisation chains that impede certification and entry to the market.

(3) Policy and Regulatory Problems

Diverse systems across the global arena create ambiguity regarding liability, long-term storage tracking, and the transportation of CO_2 across borders, reducing investment confidence. Most of the regions do not have a unified approach to CCSU, and fragmented incentives result in uneven implementation - North America and Europe are the first, and Asia considers policy gaps. The community needs to be strongly involved because the population remains unconvinced due to environmental impacts and equity issues (IPCC ,2023).

(4) Environmental and Sustainability Challenges.

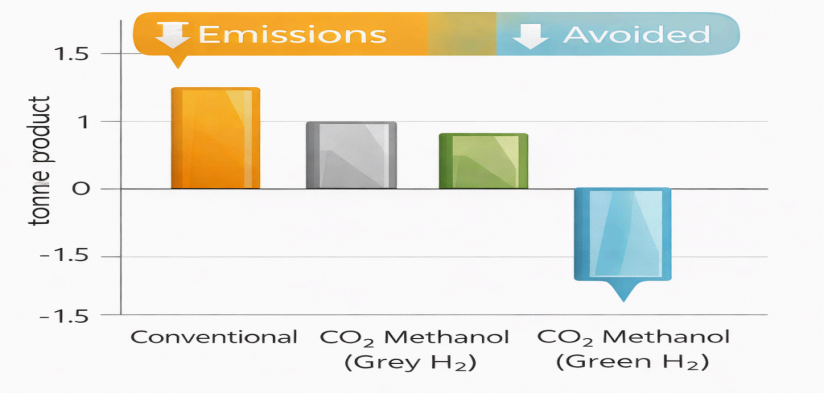
Although CCU decreases emissions, life-cycle analyses of this method indicate potential trade-offs, including increased water use in mineralization or energy-intensive processes that offset the reductions if powered by fossil fuels (IEA, 2024). Scaling should consider geographic expansion, with expansion equally distributed to prevent stress on vulnerable populations (IPCC ,2023).

Figure 5 shows Life-Cycle Carbon Analysis:

Case 1: Conventional conversion, Case 2: CO_2-to-Methanol (Grey H_2):

Net life-cycle emissions remain positive due to upstream hydrogen production

Case 3: CO_2-to-Methanol (Green H_2): Net negative emissions are possible when powered by renewable electricity



(5) Opportunities and Future Prospects.

Nonetheless, 2026 will be a milestone in the life of CCU when its market will reach USD 6.45 million and reach USD 20.4 billion at a CAGR of 13.5 by 2035 (IEA,2024). Cost and efficiency improvements will be enabled by technological innovations, such as reactive carbon capture (RCC) and chemical looping hydrogen production. Policy momentum: e.g., the EU Net-Zero Industry Act, which requires storage of 50 Mt over the five-year timeframe, and the IRA, which could unlock $80 billion in investment, could open up investment of $80 billion. Durable revenue will be generated through integration with compliance carbon markets, which represent 28% of global emissions. Carbon-to-value products such as nanotubes (a high-potential segment with 10-fold growth by 2030) and sustainable aviation fuels (another high-potential segment) are emerging uses of platinum compounds. By 2030, capture capacity may increase to 435 Mt/year, 4 times current levels, and DAC hubs such as Mammoth may reach gigaton-scale potential. The R&D of energy-efficient technology, harmonization of policies, and public-private collaboration should be the focus of the future so that the result can be a market of 70-550 billion dollars by 2030-2040.

(6) Recent Policy Developments:

India’s Investment in Carbon Capture

Recent policy developments highlight the growing global commitment toward carbon capture and utilisation technologies. In 2026, the Government of India announced a significant financial allocation of approximately USD 2.4 billion to accelerate the deployment of carbon capture, utilisation, and storage (CCSU) technologies across key industrial sectors. This initiative is aimed at facilitating large-scale implementation beyond pilot and demonstration projects, particularly in emissions-intensive industries such as steel, cement, and power generation. The funding strategy focuses on supporting innovation, infrastructure development, and commercialization of CCSU technologies, reflecting a shift from research-driven approaches to industrial-scale adoption. By prioritizing public financing and policy support, India aims to reduce emissions while maintaining industrial growth, aligning with its long-term net-zero commitments. Such initiatives indicate a broader global trend where governments are increasingly integrating CCSU into climate policy frameworks.

12. Conclusion

The use of CO_2 represents a paradigm shift, treating emissions as liabilities to the environment and using them as resources in a closed carbon economy. Synthesizing a variety of paths - chemical transformations to form methanol and urea, biotechnological transformations to produce biofuels, and mineralization to form long-lasting construction materials, this review has shown the versatility of CO_2 as the solution to climate problems and economic development. Its potential is further supported by advanced applications in the pharmaceutical sector, value-added chemical and agricultural studies, and by bibliometric trends of rapidly rising international research in hydrogenation and catalysis.

However, scaling is limited by high costs, energy requirements, and regulatory obstacles, as noted in recent studies The 2024-2026 momentum, though, growth to USD 11-22 billion by 2032-2033, investment of more than half a billion, and other projects such as Mammoth DAC are indicative of viability. By 2030, as capture capacity reaches 435 Mt/year and 2030 policies, such as the NZIA, are forced by the EU to push deployment, the CCU will mitigate 130 Mt/year in emissions (IEA ,2023) and create the desired markets of 70-550 billion by 2030-2040.

To achieve net-zero goals, stakeholders will have to focus on innovation in low-energy technologies, harmonized policies, and fair application. By using renewables and carbon credits, CO_2 emissions can guide fossil-based economies toward sustainable futures, ensuring environmental conservation and resilience in the industrial sector. In fact, coal is the major fuel of choice for much needed power generation. Coal is the cheapest fossil fuel which is abundantly available as a low cost fuel, may have to be utilized . If the cost-effective CCSU technologies become available then it may become environmentally cleaner to use coal and future generations may not have to depend only on renewable energy sources which have their own issues.

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